

RAFFLES INSTITUTION
2011 YEAR 6 PRELIMINARY EXAMINATION
Higher 1



CHEMISTRY

8872/01

Paper 1 Multiple Choice

26 September 2011

50 minutes

Additional Materials: OMR answer sheet
 Data Booklet

READ THESE INSTRUCTIONS FIRST

DO NOT open this booklet until you are told to do so.

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, class and index number in the spaces provided on the Answer Sheet.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the one you consider to be correct and record your choice with a **soft pencil** on the separate Answer Sheet.

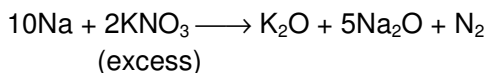
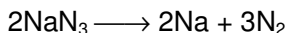
Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

Section A

For each question there are four answers, **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct and record your choice in **soft pencil** on the OMR sheet provided.

- 1 On collision, airbags in cars inflate rapidly due to the production of nitrogen gas. The nitrogen gas is formed according to the following equations.



What is the total volume of nitrogen gas produced from 1 mole of sodium azide NaN_3 , at room temperature and pressure?

- A** 36.0 dm³
B 38.4 dm³
C 72.0 dm³
D 96.0 dm³
- 2 Biochemists have discovered more than 400 mutant varieties of haemoglobin, the blood protein that carries oxygen throughout the body. A physician studying a variety of haemoglobin associated with fatal heart disease first finds its molar mass. She dissolves 21.5 mg of the protein in water at 5.0 °C to make 1.50 cm³ of solution and measures an osmotic pressure of 0.00475 atm.
- Given osmotic pressure = cRT , what is the molar mass of this haemoglobin variety?
[c = concentration in mol dm⁻³; R = 0.0821 atm dm³ mol⁻¹K⁻¹; T = temperature in K]
- A** 6.47×10^4 g mol⁻¹
B 6.89×10^4 g mol⁻¹
C 7.89×10^4 g mol⁻¹
D 8.88×10^4 g mol⁻¹
- 3 The proton number of gallium, Ga, is 31. Which of the following shows the correct order for the removal of electrons from their orbitals to form Ga⁴⁺ ion?

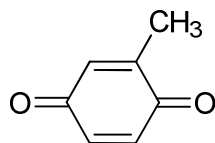
	First	Second	Third	Fourth
A	4p	4s	4s	3d
B	4p	3d	4s	4s
C	3d	4s	4s	4p
D	4s	4s	3d	4p

- 4 Which sequence of isoelectronic ions shows a continuous increasing trend of ionic

radius?

- A** $\text{Ca}^{2+} < \text{K}^+ < \text{Cl}^- < \text{S}^{2-} < \text{P}^{3-}$
B $\text{Ca}^{2+} < \text{K}^+ < \text{P}^{3-} < \text{S}^{2-} < \text{Cl}^-$
C $\text{K}^+ < \text{Ca}^{2+} < \text{Cl}^- < \text{S}^{2-} < \text{P}^{3-}$
D $\text{P}^{3-} < \text{S}^{2-} < \text{Cl}^- < \text{K}^+ < \text{Ca}^{2+}$

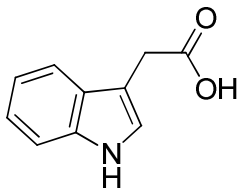
- 5** The unsaturated diketone as shown below is excreted by the bombardier beetle.



What are the numbers of sp , sp^2 and sp^3 hybridised carbon atoms in the product formed when this diketone is reacted with HCN in the presence of a trace amount of KCN ?

	sp	sp^2	sp^3
A	0	6	1
B	0	4	3
C	2	4	3
D	4	2	4

- 6** Heteroauxin is a plant hormone that plays a key role in the coordination of many growth and behavioural processes in a plant's life cycle.



heteroauxin

How many σ (sigma) and π (pi) bonds does heteroauxin have?

	σ	π
A	16	4
B	23	4
C	16	5
D	23	5

- 7 The melting point of 4-nitrophenol is higher than that of 2-nitrophenol. Which of the following best explains the above observation?
- A** The strength of hydrogen bonding is weaker in 2-nitrophenol.
- B** 4-nitrophenol has a larger surface area of contact for hydrogen bonding.
- C** There is less extensive intermolecular hydrogen bonding in 2-nitrophenol.
- D** The covalent bonds in 4-nitrophenol are stronger than that in 2-nitrophenol.

- 8** The common lead-acid car battery is a rechargeable battery and the reaction that occurs while recharging a 'dead' battery is



Given the following enthalpy changes,



Find the enthalpy change for the reaction that occurs during the recharging of the battery.

- A** +252 kJ mol⁻¹ **C** +504 kJ mol⁻¹
B +318 kJ mol⁻¹ **D** +636 kJ mol⁻¹

- 9 The enthalpy change of neutralisation of aqueous sodium hydroxide with aqueous sulfuric acid is $-57.2 \text{ kJ mol}^{-1}$, whereas the enthalpy change of neutralisation of aqueous sodium hydroxide with aqueous ethanoic acid is $-55.2 \text{ kJ mol}^{-1}$.

Which of the following best explains the numerical difference in these values?

- A** The dissociation of ethanoic acid is endothermic.
- B** A larger volume of ethanoic acid is required for completion neutralisation.
- C** Aqueous sulfuric acid has a higher concentration of hydrogen ions than ethanoic acid.
- D** Some heat energy is lost to the surrounding atmosphere for the neutralisation by ethanoic acid.

- 10** Iodine-131 is a radioactive isotope with a half-life of 8 days. Following the nuclear power plant disaster at Chernobyl in 1986, it was stated that a cloud of vapour containing iodine-131 was formed.

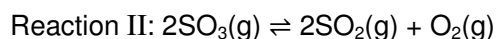
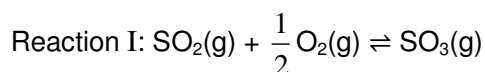
Given that radioactive decay is a first order reaction, what fraction of the isotope would remain after 80 days?

- A** $\frac{1}{20}$ **B** $\frac{1}{160}$ **C** $\frac{1}{2^8}$ **D** $\frac{1}{2^{10}}$

- 11 Which of the following correctly represents the units of the rate constant, k , for a zero order reaction?

A s^{-1}
B $\text{mol}^{-1} \text{dm}^3 \text{s}^{-1}$
C $\text{mol dm}^{-3} \text{s}^{-1}$
D $\text{mol dm}^{-3} \text{s}$

- 12 The equilibrium constant for Reaction I is K .
What is the equilibrium constant for Reaction II?



A $\frac{1}{K^2}$
B $\frac{2}{K}$
C $2K$
D K^2

- 13 Which of the following statements is correct regarding a system in dynamic equilibrium?

A All reactions have stopped.
B The concentrations of the reactants equal that of the products.
C The rate of the forward reaction equals that of the backward reaction.
D The rate constant of the forward reaction equals that of the backward reaction.

- 14 Liquid ammonia dissociates as follows:



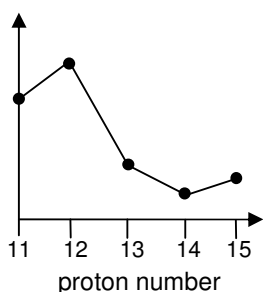
At a certain temperature, the ionic product of liquid ammonia is given as:

$$K_{\text{amm}} = [\text{NH}_4^+][\text{NH}_2^-] = 1.00 \times 10^{-22} \text{ mol}^2 \text{ dm}^{-6}$$

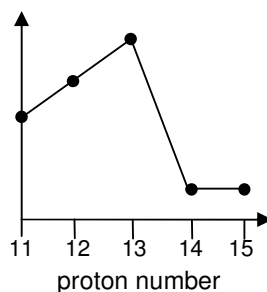
What is the total concentration of ions in this equilibrium mixture?

A $1.00 \times 10^{-11} \text{ mol dm}^{-3}$
B $2.00 \times 10^{-11} \text{ mol dm}^{-3}$
C $1.00 \times 10^{-22} \text{ mol dm}^{-3}$
D $2.00 \times 10^{-22} \text{ mol dm}^{-3}$

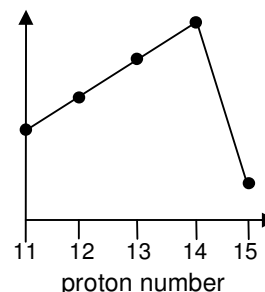
- 15 The following graphs show how three properties of the elements, Na to P, and their compounds, vary with proton number.



Graph 1



Graph 2



Graph 3

What properties are shown by the three graphs?

	Graph 1	Graph 2	Graph 3
A	Melting point of oxide	Melting point of chloride	Conductivity of element
B	Melting point of oxide	Melting point of element	Melting point of chloride
C	Melting point of chloride	Conductivity of element	Melting point of oxide
D	Melting point of element	Melting point of chloride	Conductivity of element

- 16 An element **E** has a low proton number. It forms an amphoteric oxide and a chloride which, when anhydrous, is readily hydrolysed in water.

Which Group in the Periodic Table might **E** belong to?

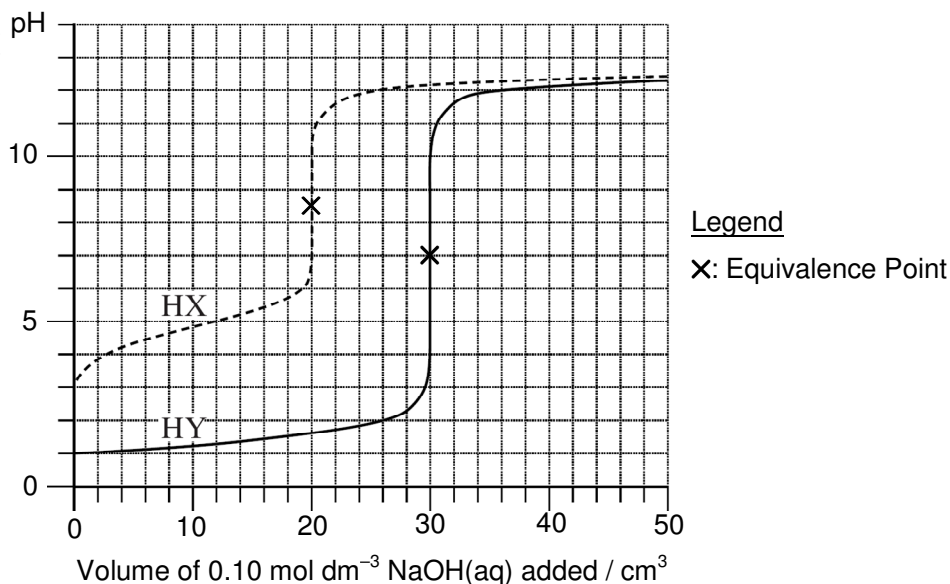
- A** II
- B** III
- C** IV
- D** V

- 17 Arsenic is below phosphorus in the Periodic Table.
Which of the following statements is **not** likely to be true?

- A** Arsenic is less electronegative than phosphorus.
- B** Liquid arsenic oxide is a good electrical conductor.
- C** An aqueous solution of arsenic chloride can turn moist blue litmus paper red.
- D** The hydride of arsenic, AsH_3 , can form an addition product with aluminium chloride.

- 18 The titration curves below were obtained using equal volumes of solutions of two

monobasic acids **HX** and **HY**.

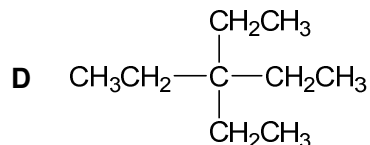
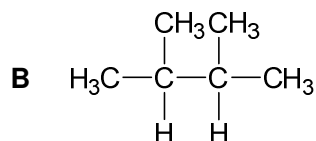
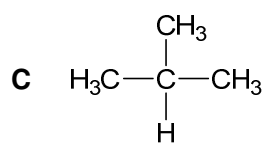
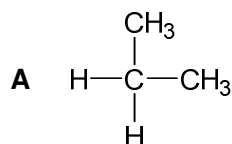


Which of the following statements about a solution of **HX** is correct?

- A** It is a weaker acid and is less concentrated than the solution of **HY**.
- B** It is a stronger acid and is less concentrated than the solution of **HY**.
- C** It is a weaker acid and is more concentrated than the solution of **HY**.
- D** It is a stronger acid and is more concentrated than the solution of **HY**.

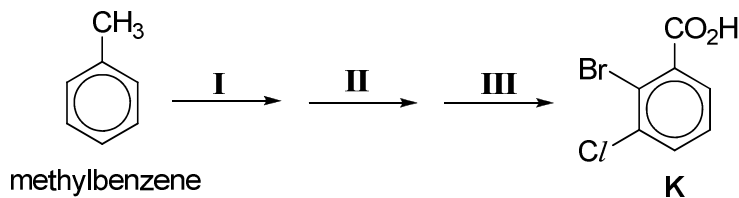
- 19** An alkane **J** is reacted with chlorine gas in the presence of ultraviolet light to form two monochlorinated alkanes in an approximate molar ratio of 6 : 1.

Which of the following is a possible structure for **J**?



- 20** Compound **K** can be synthesised from methylbenzene via a three-step synthesis

route.



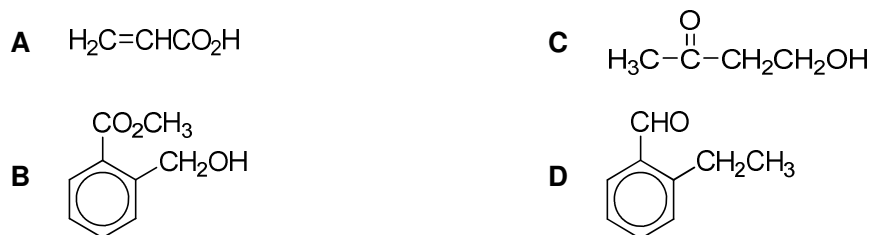
Which of the following will give the best yield for the synthesis of Compound **K** from methylbenzene?

	I	II	III
A	oxidation	chlorination	bromination
B	chlorination	oxidation	bromination
C	bromination	chlorination	oxidation
D	bromination	oxidation	chlorination

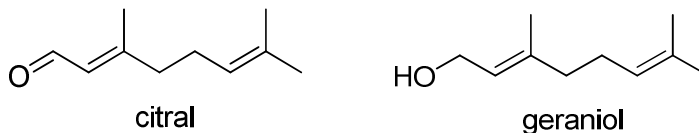
21 Compound **L** has all of the following properties:

- (i) It is neutral.
- (ii) It does not react with Fehling's solution.
- (iii) It gives an orange precipitate with 2,4-dinitrophenylhydrazine.
- (iv) It decolourises hot acidified KMnO_4 with the evolution of CO_2 gas.

What can compound **L** be?



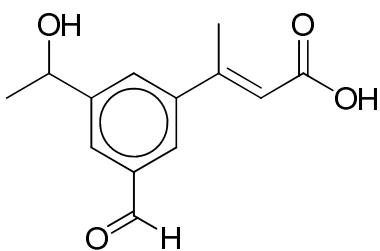
22 Citral and geraniol are chemicals commonly used in the manufacture of perfumes.



Which of the following **cannot** be used to distinguish between citral and geraniol?

- | | |
|--------------------------|---|
| A Na | C Tollens' reagent |
| B SOCl_2 | D $\text{KMnO}_4(\text{aq})$, $\text{H}_2\text{SO}_4(\text{aq})$, heat |

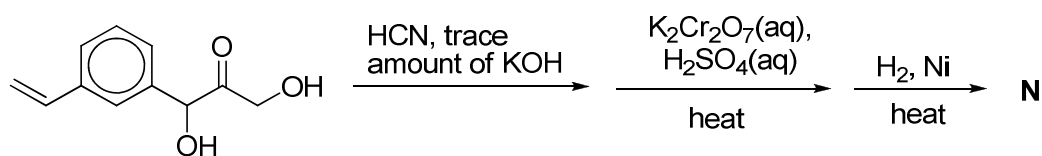
23 Compound **M** can be used as a precursor to synthesise different organic compounds under varying reagent(s) and condition.



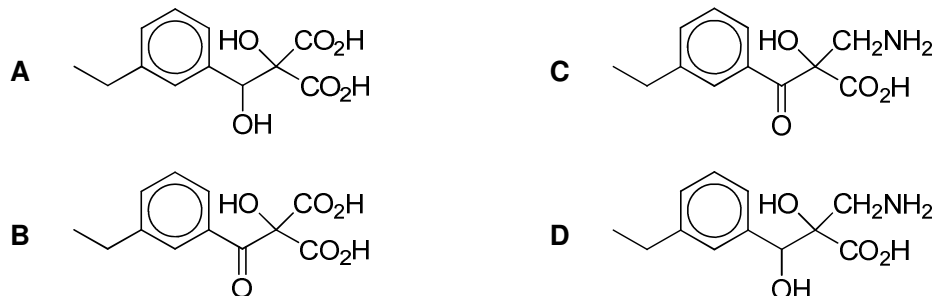
M

Which of the following shows the correct combination of the product formed under the given reagent(s) and condition?

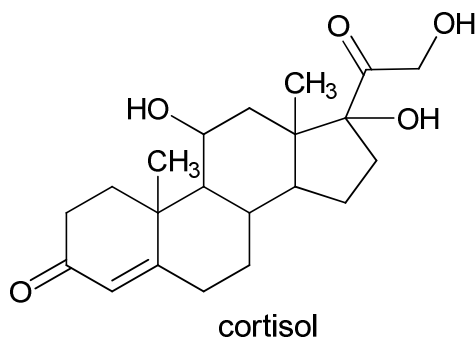
	Product	Reagent(s) and Condition
A		dry HBr(g), heat
B		K ₂ Cr ₂ O ₇ (aq), H ₂ SO ₄ (aq), heat
C		NaBH ₄ , room temperature
D		PCl ₅ , room temperature



Which of the following is the correct structure of compound **N**?



- 25** Cortisol is a steroid hormone produced by the adrenal gland.



Which of the following statements is correct?

- A** It decolourises aqueous bromine.
- B** It reacts with sodium to form 3 mol of hydrogen gas.
- C** It gives a yellow precipitate with alkaline aqueous iodine.
- D** It reacts with sodium carbonate to give carbon dioxide gas.

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements which you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 26** For which of the following pairs does species **I** have a smaller bond angle than species **II**?

	I	II
1	NO_2^-	NO_2^+
2	H_2S	SO_2
3	PH_4^+	ICl_4^-

- 27** Three colourless compounds, **P**, **Q** and **R**, react to give a coloured compound **S**. The times, *t*, taken for various initial concentrations of **P**, **Q** and **R** to produce **S** are given in the following table.

Experiment	$[\text{P}] / \text{mol dm}^{-3}$	$[\text{Q}] / \text{mol dm}^{-3}$	$[\text{R}] / \text{mol dm}^{-3}$	<i>t</i> / s
1	0.5	0.05	0.05	88
2	2.0	0.05	0.05	88
3	1.0	0.10	0.05	22
4	1.0	0.10	0.10	11

Which of the following statement(s) can be deduced from these results?

- 1** The units for the rate constant are $\text{mol}^{-1} \text{dm}^3 \text{s}^{-1}$.
- 2** The rate equation for the reaction is $\text{rate} = k[\text{Q}]^2[\text{R}]$.
- 3** The half-life of the reaction with respect to **R** remains the same when the concentration of **R** is doubled.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

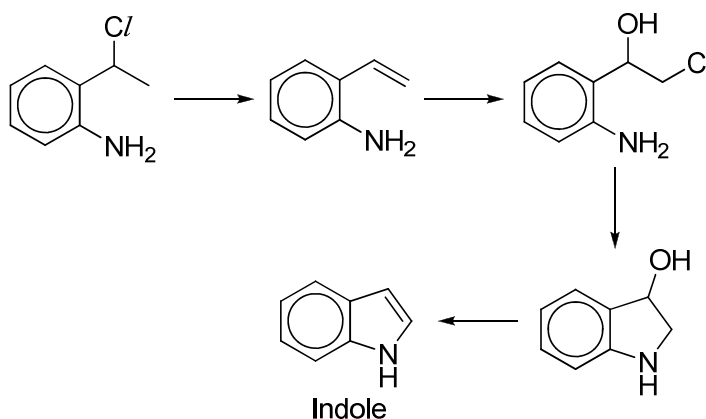
No other combination of statements is used as a correct response.

- 28** The equilibrium constant, K_c , for the reaction $\mathbf{V(aq)} \rightleftharpoons \mathbf{W(aq)}$ is determined at various temperatures.

Temperature, K	Equilibrium constant, K_c
283	0.142
293	0.142
303	0.142

Which of the following statement(s) can be deduced from these results?

- 1 The reaction occurs without any heat changes.
 - 2 The forward and backward reactions have the same activation energy.
 - 3 The forward and backward reactions have the same overall order of reactions.
- 29** Indole is an organic compound found in natural jasmine oil and is the precursor to many pharmaceuticals. The diagram below shows a reaction scheme to synthesize indole.



Which of the following types of reaction are involved in the above reaction scheme?

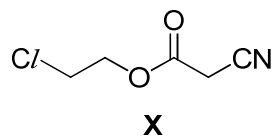
- 1 condensation
- 2 addition
- 3 substitution

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 30** Which of the following will be formed when compound **X** is boiled with aqueous sodium hydroxide?



- 1**
- 2**
- 3**

END OF PAPER

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